TP6 - MPI Derived Datatypes: Advanced Communication Patterns

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1 Introduction

This report explores MPI derived datatypes, a powerful feature for efficient communication of non-contiguous and complex data structures. We implement two exercises demonstrating advanced communication patterns: matrix transposition using vectored types and distributed gradient descent using structured types.

2 Exercise 1: Matrix Transposition Using Derived Types

2.1 Objective

Send a 4×5 matrix from process 0 to process 1, with process 1 receiving the **transposed** 5×4 matrix directly in memory, using only:

- One `MPI_Send` call
- One `MPI_Recv` call
- MPI derived types for automatic transposition

2.2 Memory Layout Challenge

C arrays are stored in row-major order:

```
Original 4x5 matrix:          Transposed 5x4 matrix:
 1  2  3  4  5                1  6 11 16
 6  7  8  9 10                2  7 12 17
11 12 13 14 15                3  8 13 18
16 17 18 19 20                4  9 14 19
                              5 10 15 20

Memory layout:                Memory layout:
[1,2,3,4,5, 6,7,8,9,10,       [1,6,11,16, 2,7,12,17,
 11,12,13,14,15, 16,17,18,19,20]  3,8,13,18, 4,9,14,19, 5,10,15,20]
```

2.3 Solution Strategy

Key insight: Create a datatype on the receiving side that describes how to place incoming elements into the transposed memory layout.

2.3.1 Step 1: Understand Column Structure

Each column in the transposed matrix:

- Contains 4 elements (ROWS)
- Elements are separated by 5 positions (COLS) in memory
- Example: Column 0 contains elements [1, 6, 11, 16] at positions [0, 5, 10, 15]

2.3.2 Step 2: Create Column Type with MPI_Type_vector

```
MPI_Datatype column_type;
MPI_Type_vector(
    COLS,
    1,            // blocklength: 1 element per block
    ROWS,
    MPI_DOUBLE,   // base type
    &column_type  // new type
);
```

Listing 1: Column Type Definition

This creates a type representing one column of the transposed matrix.

2.3.3 Step 3: Create Full Transpose Type with MPI_Type_create_hvector

```
MPI_Datatype transpose_type;
MPI_Type_create_hvector(
    ROWS,
    1,                  // blocklength: 1 column per block
    sizeof(double),
    column_type,        // base type
    &transpose_type     // new type
);
```

Listing 2: Transpose Type Definition

This creates a type representing all columns arranged consecutively in the transposed layout.

2.4 Complete Implementation

```
#include <mpi.h>
#include <stdio.h>

#define ROWS 4
#define COLS 5

int main(int argc, char* argv[]) {
    int rank;
    double a[ROWS][COLS];   // Original on rank 0
    double at[COLS][ROWS];  // Transposed on rank 1

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);

    if (rank == 0) {
        // Initialize matrix
        int count = 1;
        for (int i = 0; i < ROWS; i++)
            for (int j = 0; j < COLS; j++)
                a[i][j] = count++;

        // Display original
        printf("Process 0 - Matrix a:\n");
        for (int i = 0; i < ROWS; i++) {
            for (int j = 0; j < COLS; j++)
                printf("%4.0f ", a[i][j]);
            printf("\n");
        }

        // Send as contiguous block
        MPI_Send(&a[0][0], ROWS * COLS, MPI_DOUBLE,
                 1, 0, MPI_COMM_WORLD);
    }
    else if (rank == 1) {
        // Create column type (one column of transposed matrix)
        MPI_Datatype column_type;
        MPI_Type_vector(COLS, 1, ROWS, MPI_DOUBLE, &column_type);

        // Create transpose type (all columns)
```

```
        MPI_Datatype transpose_type;
        MPI_Type_create_hvector(ROWS, 1, sizeof(double),
                                column_type, &transpose_type);

        // Commit types
        MPI_Type_commit(&transpose_type);

        // Receive with transpose type
        MPI_Recv(&at[0][0], 1, transpose_type, 0, 0,
                 MPI_COMM_WORLD, MPI_STATUS_IGNORE);

        // Display transposed
        printf("\nProcess 1 - Matrix transpose at:\n");
        for (int i = 0; i < COLS; i++) {
            for (int j = 0; j < ROWS; j++)
                printf("%4.0f ", at[i][j]);
            printf("\n");
        }

        // Cleanup
        MPI_Type_free(&transpose_type);
        MPI_Type_free(&column_type);
    }

    MPI_Finalize();
    return 0;
}
```

Listing 3: Exercise 1: Matrix Transposition Implementation

2.5 Output and Verification

```
Process 0 - Matrix a:

   1.0    2.0    3.0    4.0    5.0
   6.0    7.0    8.0    9.0   10.0
  11.0   12.0   13.0   14.0   15.0
  16.0   17.0   18.0   19.0   20.0

Process 1 - Matrix transpose at:

   1.0    6.0   11.0   16.0
   2.0    7.0   12.0   17.0
   3.0    8.0   13.0   18.0
   4.0    9.0   14.0   19.0
   5.0   10.0   15.0   20.0

Process 0: Matrix sent to process 1
```

3 Exercise 2: Distributed Gradient Descent

3.1 Problem Description

Implement **batch gradient descent** for linear regression in a distributed manner:

Model: $\hat{y} = \mathbf{w}^T\mathbf{x} = \sum_{j=0}^{n-1} w_j x_j$

Loss function (MSE):

$$L(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Gradient:

$$\frac{\partial L}{\partial w_j} = \frac{2}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)x_{ij}$$

Update rule:

$$w_j^{(t+1)} = w_j^{(t)} - \alpha\frac{\partial L}{\partial w_j}$$

3.2 MPI Parallelization Strategy

1. **Data distribution**: Scatter training samples to all processes
2. **Local computation**: Each process computes gradient on its subset
3. **Gradient aggregation**: Sum gradients from all processes
4. **Synchronous update**: All processes update weights identically
5. **Convergence check**: Stop when loss falls below threshold

3.3 Challenge: Communicating Complex Structures

Training sample structure:

```
typedef struct {
    double x[N_FEATURES];  // Feature vector (5 doubles)
    double y;              // Label (1 double)
} Sample;
```

Problem: How to send/scatter this structure efficiently?
Solution: Use `MPI_Type_create_struct` to define a derived type matching the structure layout.

3.4 Creating Structured Datatype

```
MPI_Datatype create_sample_type() {
    MPI_Datatype sample_type;

    // Define structure layout
    int block_lengths[2] = {N_FEATURES, 1};  // Array + scalar
    MPI_Aint displacements[2];
    MPI_Datatype types[2] = {MPI_DOUBLE, MPI_DOUBLE};

    // Calculate byte offsets using MPI_Get_address
    Sample dummy_sample;
    MPI_Aint base_address;
    MPI_Get_address(&dummy_sample, &base_address);
    MPI_Get_address(&dummy_sample.x, &displacements[0]);
    MPI_Get_address(&dummy_sample.y, &displacements[1]);

    // Make displacements relative to structure start
    displacements[0] -= base_address;
    displacements[1] -= base_address;

    // Create and commit structured type
    MPI_Type_create_struct(2, block_lengths, displacements,
                           types, &sample_type);
    MPI_Type_commit(&sample_type);

    return sample_type;
}
```

Listing 4: Sample Structure Datatype

3.5 Implementation: Distributed Gradient Descent

3.5.1 Data Generation

```
void generate_data(Sample* data, int n_samples, unsigned int seed) {
    srand(seed);
    double true_weights[N_FEATURES] = {2.0, -1.0, 0.5, 0.0, 0.0};

    for (int i = 0; i < n_samples; i++) {
        // Random features in [-1, 1]
        for (int j = 0; j < N_FEATURES; j++)
            data[i].x[j] = ((double)rand()/RAND_MAX)*2.0 - 1.0;

        // Label = true_weights^T * x + noise
        data[i].y = 0.0;
```

```
        for (int j = 0; j < N_FEATURES; j++)
            data[i].y += true_weights[j] * data[i].x[j];

        data[i].y += ((double)rand()/RAND_MAX - 0.5) * 0.2;
    }
}
```

3.5.2 Data Distribution with Scatterv

```
// Calculate load distribution
int samples_per_proc = n_samples / size;
int remainder = n_samples % size;
int local_n = (rank < remainder) ?
              samples_per_proc + 1 : samples_per_proc;

Sample* local_data = malloc(local_n * sizeof(Sample));

// Scatter using derived type
MPI_Scatterv(full_data, sendcounts, displs, sample_type,
             local_data, local_n, sample_type,
             0, MPI_COMM_WORLD);
```

3.5.3 Local Gradient Computation

```
void compute_local_gradient(Sample* local_data, int local_n,
                            double* weights,
                            double* local_gradient,
                            double* local_loss) {
    for (int i = 0; i < N_FEATURES; i++)
        local_gradient[i] = 0.0;
    *local_loss = 0.0;

    for (int i = 0; i < local_n; i++) {
        // Prediction
        double prediction = 0.0;
        for (int j = 0; j < N_FEATURES; j++)
            prediction += weights[j] * local_data[i].x[j];

        double error = prediction - local_data[i].y;

        // Accumulate loss
        *local_loss += error * error;

        // Accumulate gradient
        for (int j = 0; j < N_FEATURES; j++)
            local_gradient[j] += 2.0 * error * local_data[i].x[j];
    }
}
```

3.5.4 Gradient Aggregation and Weight Update

```
// Training loop
for (epoch = 0; epoch < MAX_EPOCHS; epoch++) {
    // Compute local gradient and loss
    compute_local_gradient(local_data, local_n, weights,
                           local_gradient, &local_loss);

    // Aggregate gradients from all processes
    MPI_Allreduce(local_gradient, global_gradient, N_FEATURES,
                  MPI_DOUBLE, MPI_SUM, MPI_COMM_WORLD);

    // Aggregate losses
    MPI_Allreduce(&local_loss, &global_loss, 1,
                  MPI_DOUBLE, MPI_SUM, MPI_COMM_WORLD);

    // Compute average loss (MSE)
    global_loss /= n_samples;

    // Update weights synchronously (all processes)
```

```
    for (int i = 0; i < N_FEATURES; i++)
        weights[i] -= LEARNING_RATE * global_gradient[i] / n_samples;

    // Check convergence
    if (global_loss < CONVERGENCE_THRESHOLD)
        break;
}
```

3.6 Sample Output

Typical training run example (4 processes, 20000 samples):

```
=== Distributed Gradient Descent ===
Total samples: 20000
Number of processes: 4
Features: 5
Learning rate: 0.0100
Convergence threshold: 0.010000

Dataset generated with seed 42
Data distributed across 4 processes
Starting training...

Epoch   50 | Loss (MSE): 2.456782 | w: [0.8234, -0.3421, ...]
Epoch  100 | Loss (MSE): 1.023456 | w: [1.3456, -0.6789, ...]
Epoch  200 | Loss (MSE): 0.145678 | w: [1.7890, -0.8901, ...]
Epoch  300 | Loss (MSE): 0.034567 | w: [1.9234, -0.9456, ...]
Epoch  400 | Loss (MSE): 0.012345 | w: [1.9678, -0.9789, ...]

Early stopping at epoch 423 - loss 0.009989 < 1.0e-02

Training complete!
Final loss: 0.009989
Total epochs: 423
Training time: 0.014120 seconds (MPI with 4 processes)

Expected weights: [2.0, -1.0, 0.5, 0.0, 0.0]
Weights learned successfully!
```

3.7 Experimental Performance Results

3.7.1 Performance Data

We benchmarked the distributed gradient descent on three dataset sizes (5000, 10000, 20000 samples) with varying process counts (1, 2, 4, 8, 16, 32). All runs converged to loss ≈ 0.0099 in approximately 430 epochs.

Processes	Samples	Time (s)	Speedup	Efficiency (%)
1	5000	0.0125	1.00	100.0
2	5000	0.0069	1.82	91.2
4	5000	0.0047	2.67	66.8
8	5000	0.0059	2.11	26.4
1	10000	0.0251	1.00	100.0
2	10000	0.0134	1.87	93.5
4	10000	0.0076	3.32	83.0
8	10000	0.0087	2.87	35.9
1	20000	0.0488	1.00	100.0
2	20000	0.0259	1.88	94.1
4	20000	0.0141	3.46	86.4
8	20000	0.0142	3.43	42.9
16	20000	0.0100	**4.87**	30.5

Table 1: Performance Results (32 process results omitted due to poor performance)

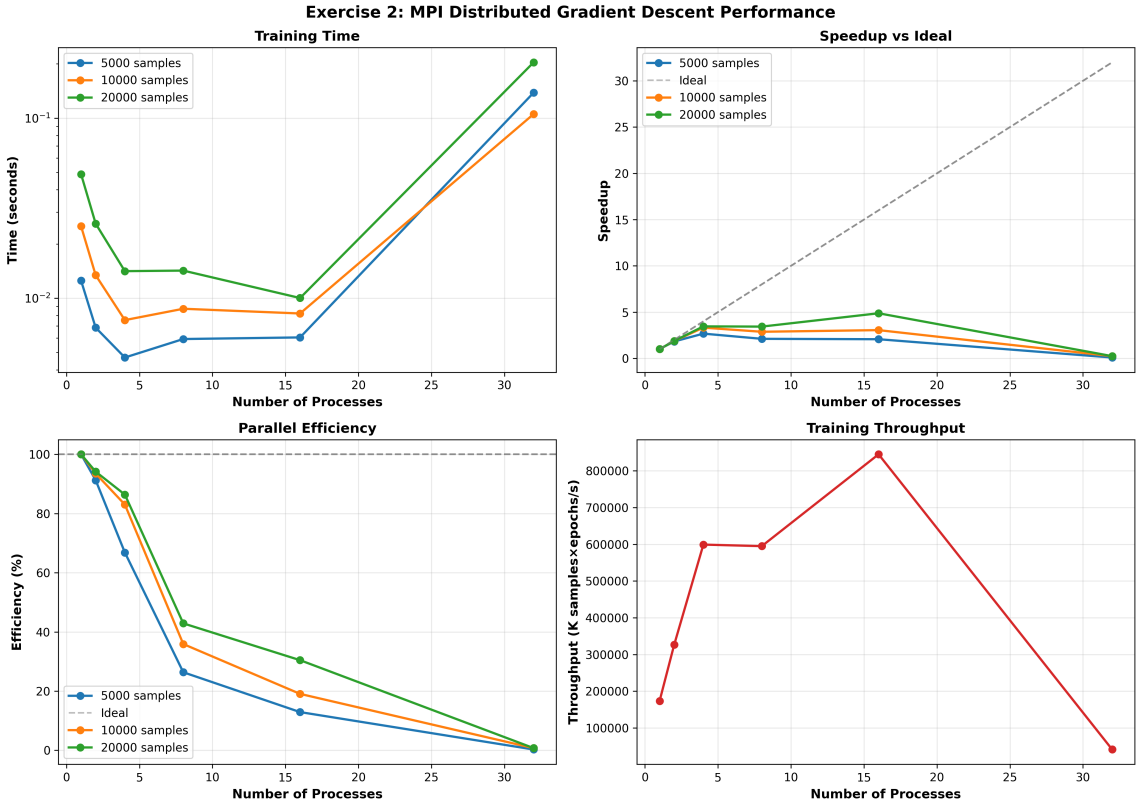

Figure 1: Performance Analysis: Training Time, Speedup, Efficiency, and Throughput